



The Islamia University of Bahawalpur

E-Class Room

**A project presented to
Department of Information Technology, IUB Bahawalpur**

**In partial fulfillment
of the requirement for the degree of**

Bachelor of Science in Information Technology (2022-2026)

By

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F22BINFT1E02024

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Student Name
Kanwal Shahzadi

CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (IT) “E-Class”
was developed by K a n w a l S h a h z a d i ,F22BINFT1E02024, 2022
-2026 under the supervision of “Ms. Abdul Kainat ”and that in his opinion;
it is fully adequate, in scope and quality for the degree of Bachelors of Science in
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Acknowledgement

I would like to express my sincere gratitude to everyone who supported me throughout the course of this project. Special thanks to my supervisor for their valuable guidance and continuous encouragement. I am deeply grateful to my family and friends for their unwavering support and motivation. My heartfelt appreciation also goes to my peers for their insightful feedback and assistance. I would also like to acknowledge all the resources and tools that enabled me to successfully complete this project. Without their collective contributions, this work would not have been possible.

Student Name

Sayyam Ali

Abbreviations

Provide a list of all abbreviation used in the document.

SRS	Software Require Specification
PC	Personal Computer
SDD	Software Design Document

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1 Introduction

This chapter introduces the E-Class app, an online school attendance tracking system designed to simplify attendance management for teachers and enhance communication with parents. The app aims to streamline the process by allowing teachers to efficiently record student attendance, while parents can stay informed about their child's status in real-time. The user-friendly interface ensures ease of use, while features such as manual and automated attendance tracking, reporting, and notifications ensure an efficient and seamless experience. This chapter outlines the core functionalities of the app, including user registration, attendance tracking, reporting, and the notification manager. The overall goal is to improve attendance management, accountability, and communication in the school environment.

1.1 Brief

The E-Class app is an online attendance tracking system designed to simplify the attendance management process for teachers and enhance communication with parents. The main outcome of this project is the development of a user-friendly platform that allows teachers to record and manage student attendance, while parents receive real-time updates on their child's attendance status. The app is built using HTML, CSS, JavaScript, React, and a default JavaScript backend for the backend operations. Key methodologies include agile development, focusing on user experience and efficient data handling. The report discusses the features of the app, including user registration, attendance tracking, reporting, and notifications, highlighting their functionalities and technical implementation in separate chapters.

1.2 Relevance to Course Modules

This project is closely related to several key courses studied during my Bachelor's in Software Engineering. The development of the E-Class app draws upon knowledge gained from courses such as **Web Development**, where I learned to design and implement user interfaces using HTML, CSS, and JavaScript. The **Object-Oriented Software Engineering (OOSE)** course provided a foundation for structuring the application's backend logic and ensuring maintainable and modular code. Additionally, concepts from

Database Management were applied in storing and managing attendance records securely using a backend system. The app's design and implementation also align with **IT Project Management** principles, as it involved planning, development, and testing phases, ensuring timely and effective project delivery.

1.3 Project Background

The idea behind the E-Class app is to address the challenges faced by schools in efficiently managing student attendance and improving communication with parents. Traditionally, attendance tracking is done manually, which can be time-consuming and prone to errors. The E-Class app aims to streamline this process by providing a digital platform where teachers can easily mark student attendance, and parents can receive real-time updates regarding their child's attendance status.

The app allows teachers to track student attendance, either manually or through automated methods, such as biometric systems or RFID, offering flexibility and accuracy in data collection. The system also generates detailed reports on attendance, which can be customized based on parameters such as date range or student name. Additionally, the Notification Manager feature ensures that parents are immediately notified of any attendance changes, fostering better communication and accountability. By digitizing the attendance process, the E-Class app enhances the overall school experience for both teachers and parents, making attendance management more efficient and transparent.

1.4 Related Material and Literature

The E-Class app aligns with current trends in educational technology, particularly in the area of **attendance management systems**. Many schools and educational institutions are increasingly adopting digital tools to automate administrative tasks, and attendance tracking is no exception. Research shows that automated attendance systems can improve accuracy, save time, and reduce administrative costs.

Biometric and RFID-based Attendance Systems: One of the major advancements in attendance tracking is the use of biometric systems or RFID technology, which enhances accuracy and prevents proxy attendance. Products such as **ZK Teco** and **Aadhar-enabled**

biometric systems are examples of how biometric data is used for attendance tracking in educational settings. These technologies are integrated into some educational software systems to improve both efficiency and security.

Cloud-based Attendance Systems: With the rise of cloud computing, many institutions are opting for cloud-based attendance solutions to ensure scalability and data security. Products like **Google Classroom** and **Moodle** provide attendance tracking as part of a broader suite of tools aimed at digital learning management. These systems often feature real-time data syncing, enabling teachers and parents to stay updated on attendance statuses.

Mobile Attendance Apps: The increasing use of smartphones has led to the development of mobile-based attendance apps. Solutions such as **Attendify** and **MyAttendanceTracker** enable teachers to mark attendance and send notifications directly from their mobile devices. This trend reflects the growing need for portability and real-time communication between educators and parents.

The E-Class app builds upon these trends by offering a **web-based attendance management system** that integrates real-time notifications and customizable reporting features, making it an innovative solution for modern-day schools. The research highlights the need for such systems to increase efficiency and foster better communication between educators and parents, making the E-Class app relevant in today's education landscape.

1.5 Analysis from Literature Review

In comparing the E-Class app with the trends and products discussed in the literature review, several key insights emerge that highlight the strengths and unique aspects of our project.

1. **Automated Attendance Systems:** Literature on automated attendance systems emphasizes the accuracy and efficiency brought by technologies such as biometrics and RFID. While these systems often require specialized hardware, E-Class takes a more

flexible approach by allowing manual and automated attendance recording, catering to a wider range of schools with varying resources. This adaptability is an advantage over existing systems that may be restricted to a specific type of technology.

2. **Cloud-based Attendance Solutions:** Research on cloud-based attendance solutions shows a growing preference for scalable, accessible platforms. However, many of these solutions are integrated into larger learning management systems (LMS) and may not offer the specialized focus on attendance alone. E-Class, in contrast, is dedicated to the attendance process, making it more streamlined for users who need a focused tool without additional complexity from other educational features.
3. **Mobile Attendance Apps:** Mobile apps for attendance tracking are increasingly popular, reflecting a trend towards portability and real-time communication. While platforms like **Attendify** provide mobile-based solutions, E-Class sets itself apart by not only offering mobile-friendly interfaces but also by providing a robust notification system that updates parents instantly. This real-time communication, while common in mobile apps, is a central feature in E-Class, as it aligns with the growing demand for parents to be continuously informed about their child's progress.
4. **Reporting and Data Analytics:** Many existing systems generate reports based on attendance data but lack customization in terms of formatting and filtering. The E-Class app addresses this gap by offering comprehensive reporting tools that allow users to generate attendance reports in various formats (PDF, Excel, CSV) and filter them based on multiple criteria, such as student grade level or specific attendance statuses (e.g., tardy, absent). This level of customization is less commonly seen in the market, providing added value to schools and educators.
5. **User Experience and Interface Design:** The user-centric design and intuitive interface of E-Class align well with trends in modern educational software, where ease of use is crucial. While many platforms attempt to balance features with usability, E-Class places a strong emphasis on simplicity and ease of navigation, making it accessible to a wider range of users, from teachers to parents, without extensive training.

In conclusion, the E-Class app stands out in several key areas: its flexibility in attendance tracking, real-time notifications, customizable reporting, and user-friendly design. While literature shows various trends in the attendance management space, E-Class's unique combination of features positions it as a highly adaptable and accessible solution for schools looking to modernize their attendance systems.

1.6 Methodology and Software Lifecycle for This Project

For the development of the E-Class app, we adopted an **Agile methodology** to ensure flexibility, adaptability, and continuous improvement throughout the project. Agile methodologies are ideal for projects that require frequent updates, iterations, and close collaboration between developers and stakeholders. This approach allowed us to work incrementally, focusing on delivering working features in short cycles (sprints), while adapting to changing requirements and feedback.

Software Development Lifecycle (SDLC)

We followed the **Iterative and Incremental model** of the Software Development Lifecycle (SDLC) for this project. This model is well-suited for projects where requirements may evolve over time, and the development process is broken into smaller, manageable phases.

1. **Requirement Gathering and Analysis:** The first phase involved understanding the problem, defining the scope of the app, and gathering functional and non-functional requirements from stakeholders (teachers, parents, and school administrators). We also analyzed existing systems to understand their limitations and incorporate improvements into the E-Class app.
2. **System Design:** In this phase, we focused on creating the system architecture, defining the app's user interface (UI) using wireframes and prototypes, and planning the backend logic. We also decided on the tech stack: HTML, CSS, JavaScript, React for the frontend, and a default JavaScript backend for the server-side logic.
3. **Implementation:** The actual coding and development phase followed, where the frontend and backend were built incrementally. Features such as user registration, attendance tracking, reporting, and notifications were developed and tested in separate iterations. The agile approach allowed us to refine features based on user feedback and progress with each sprint.

4. **Testing:** As part of the iterative process, testing was performed continuously throughout the project. We conducted unit testing for individual features, integration testing to ensure all components worked together, and user acceptance testing (UAT) with stakeholders to ensure the app met their needs. The app's usability and performance were prioritized to ensure smooth user experience.
5. **Deployment:** After final testing, the app was deployed to a staging environment where further testing was conducted before the live launch. We ensured the system was scalable and secure, using appropriate security measures for user data.
6. **Maintenance and Updates:** Post-deployment, we adopted an ongoing maintenance process where the system is regularly updated based on feedback from users and evolving requirements. New features, bug fixes, and performance improvements will continue to be added in future sprints.

In conclusion, the combination of Agile methodology and the Iterative and Incremental SDLC model allowed us to remain flexible and responsive to changes during the development of the E-Class app. This ensured the project was delivered on time, met user expectations, and is well-positioned for future enhancements.

1.6.1 Rationale behind Selected Methodology

The **Agile methodology** and **Iterative and Incremental SDLC model** were selected for the E-Class app due to their flexibility, adaptability, and suitability for developing dynamic, user-focused applications. Below is a detailed rationale for selecting these methodologies and SDLC model:

Agile Methodology

1. **Flexibility and Adaptability:** The Agile methodology is ideal for projects like E-Class, where requirements may evolve during development. Agile allows for quick adjustments and refinements based on user feedback, which is essential for an app focused on user experience and real-time communication. As the project involves both backend and frontend components, the ability to adapt to new requirements or changes in features without disrupting the entire project was crucial.

2. **Iterative Development:** Agile encourages working in **sprints** and delivering small, incremental features, which means the development team can quickly see progress and get feedback from stakeholders. This is important in a school-based app like E-Class, where educators and parents can provide input at regular intervals to ensure the app meets their needs. The iterative nature ensures that the final product is continually refined based on real-world usage.
3. **Collaboration and Communication:** Since the project involved multiple stakeholders, including teachers and parents, Agile methodology fosters regular collaboration between developers and users. It ensures that the development team remains in close communication with stakeholders throughout the development process, helping ensure the product delivers value to the end-users.

Iterative and Incremental SDLC Model

1. **Continuous Improvement:** The Iterative and Incremental model allows for building the system in small, manageable sections while refining the product with each iteration. This approach was selected because it enables the E-Class app to be released in stages, with each version adding new features or improving existing ones. The focus on incremental development allowed for early testing and feedback to identify potential issues early in the process.
2. **Handling Evolving Requirements:** The Iterative and Incremental model is particularly useful for handling evolving requirements, which is often the case with user-centric applications. As the E-Class app aims to meet the specific needs of both teachers and parents, their expectations and requirements may change as the app is developed. This model allows for addressing those changes without overhauling the entire project.
3. **Risk Management:** By developing the system incrementally, risks can be identified and addressed early on in each iteration. This ensures that problems, such as bugs or performance issues, can be corrected before they escalate, leading to a more stable and reliable final product.

Why Not Structural or Object-Oriented Methodologies?

- **Structural Methodology:** A more rigid, sequential approach like the Waterfall model (a structural methodology) might not have been suitable for this project because it does not allow for significant changes during development. Given that user feedback is critical for E-Class's success, a flexible, iterative approach was needed.

- **Object-Oriented Methodology:** While Object-Oriented Programming (OOP) is important in the development process, especially for creating reusable code, the project's focus was more on delivering features rapidly and adapting based on feedback. The Agile approach allowed us to break down features into manageable tasks and develop them efficiently, while Object-Oriented principles were applied during the design and implementation phases to ensure maintainability and scalability of the code.

In conclusion, the Agile methodology and Iterative and Incremental SDLC model were chosen for the E-Class app because they best align with the dynamic nature of the project, ensuring flexibility, frequent user feedback, and the continuous improvement of the system based on real-time needs.

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- *Iterative Development:* Agile encourages working in *sprints* and delivering small, incremental features, which means the development team can quickly see progress and get feedback from stakeholders. This is important in a school-based app like E-Class, where educators and parents can provide input at regular intervals to ensure the app meets their needs. The iterative nature ensures that the final product is continually refined based on real-world usage.

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2 Problem Definition

The primary problem that the E-Class app aims to solve is the inefficiency and inaccuracy in traditional student attendance tracking methods used in schools. In many educational institutions, attendance is still recorded manually, either on paper or in spreadsheets, which can be time-consuming, prone to human error, and difficult to manage. Additionally, current systems often fail to provide real-time updates to parents, limiting communication and transparency regarding their child's attendance status.

Another issue is the lack of customization in attendance reporting systems. Many existing systems generate basic reports that do not allow educators or administrators to filter data based on specific parameters such as attendance status or student grade level. This limits the ability of teachers and parents to easily track and analyze attendance patterns.

Outcome

The E-Class app provides a solution to these challenges by offering a digital, real-time attendance tracking system that can be accessed by teachers, students, and parents. The app enables teachers to efficiently mark attendance, either manually or through automated systems like RFID or biometric scanning, and it stores the data securely. It also allows for the generation of customizable reports in various formats (e.g., PDF, Excel, CSV), which can be filtered based on specific criteria, making attendance data more actionable and useful.

Additionally, the app ensures that parents are notified instantly about their child's attendance status, fostering better communication between schools and families. The outcome of this project is the creation of a streamlined, efficient, and user-friendly solution to attendance management, which benefits not only teachers and parents but also the students themselves by providing greater accountability and transparency.

2.1 Problem Statement

The traditional methods of student attendance tracking, such as manual entry or spreadsheet-based systems, are often inefficient and prone to human errors. These methods lead to inaccurate records, time-consuming processes for teachers, and delayed or inadequate communication with parents. As a result, both teachers and parents face challenges in keeping track of student attendance in real-time, which can impact student performance and accountability. Additionally, the lack of customization in attendance reporting systems makes it difficult for educators to analyze attendance patterns or generate meaningful insights.

There is a clear need for an automated, user-friendly, and customizable attendance management solution that can streamline the process, improve communication between parents and schools, and provide actionable attendance data to all stakeholders. The E-Class app aims to solve this problem by offering a digital platform that allows teachers to easily record attendance, parents to receive real-time updates, and administrators to generate customized reports with ease.

2.2 Deliverables and Development Requirements

Deliverables:

1. **E-Class App (Final Product):** The main deliverable is the fully functional E-Class app, which includes all features related to attendance tracking, reporting, user registration, and notification management. The app will be accessible to teachers, parents, and administrators, providing real-time attendance updates and easy-to-generate reports.
2. **Documentation:**
 - **User Manual:** A comprehensive guide explaining how to use the E-Class app for teachers, parents, and administrators. It will cover features such as marking attendance, generating reports, managing notifications, and handling user accounts.
 - **Technical Documentation:** Detailed documentation of the code, system architecture, database schema, and APIs (if applicable) for future developers to maintain and enhance the app.

3. **Source Code:** The complete source code of the E-Class app, including both the frontend (HTML, CSS, JavaScript, React) and the backend (JavaScript backend), will be delivered for future development or modification.
4. **Testing Reports:** Documentation of all testing activities, including unit tests, integration tests, user acceptance testing (UAT), and bug fixes. This will include results, issues encountered, and how they were resolved during the development phase.
5. **Deployment and Maintenance Plan:** A plan for deploying the app to a live environment, along with details on its future maintenance, updates, and security measures.

Development Requirements:

1. Technical Requirements:

- **Frontend:** The app should be developed using HTML, CSS, JavaScript, and React for a responsive and user-friendly interface.
- **Backend:** The backend will be developed using JavaScript (Node.js or another JavaScript-based framework), which will handle user authentication, attendance data management, and real-time notifications.
- **Database:** A secure, lightweight database such as SQLite or a cloud-based database (depending on the scale and deployment) will be used to store user and attendance data.
- **Authentication:** Implement secure user authentication, with roles such as teacher, parent, and admin, ensuring appropriate access to different features of the app.
- **Notification System:** Implement a real-time notification system to alert parents about their child's attendance status.

2. Functional Requirements:

- **User Registration and Login:** Users must be able to register, log in, and manage their accounts. Email verification for account activation should be implemented.
- **Attendance Tracking:** Teachers must be able to mark attendance as present, absent, or tardy for students in each class. Attendance should be stored with a timestamp and be editable if needed.
- **Report Generation:** Users should be able to generate attendance reports in different formats (PDF, Excel, CSV) with the ability to filter and sort data.
- **Notification Management:** Notifications should be sent to parents in real-time when their child's attendance status changes (e.g., absent, tardy).

3. Non-Functional Requirements:

- **Security:** Ensure data security and privacy, particularly for sensitive information like student attendance records.
- **Scalability:** The app should be designed to scale efficiently as the number of users increases.

- **Usability:** The user interface should be intuitive and easy to navigate, ensuring that all users (teachers, parents, admins) can interact with the app without extensive training.
- **Performance:** The app should load quickly and handle multiple users accessing the system simultaneously without performance degradation.

4. **Deployment Requirements:**

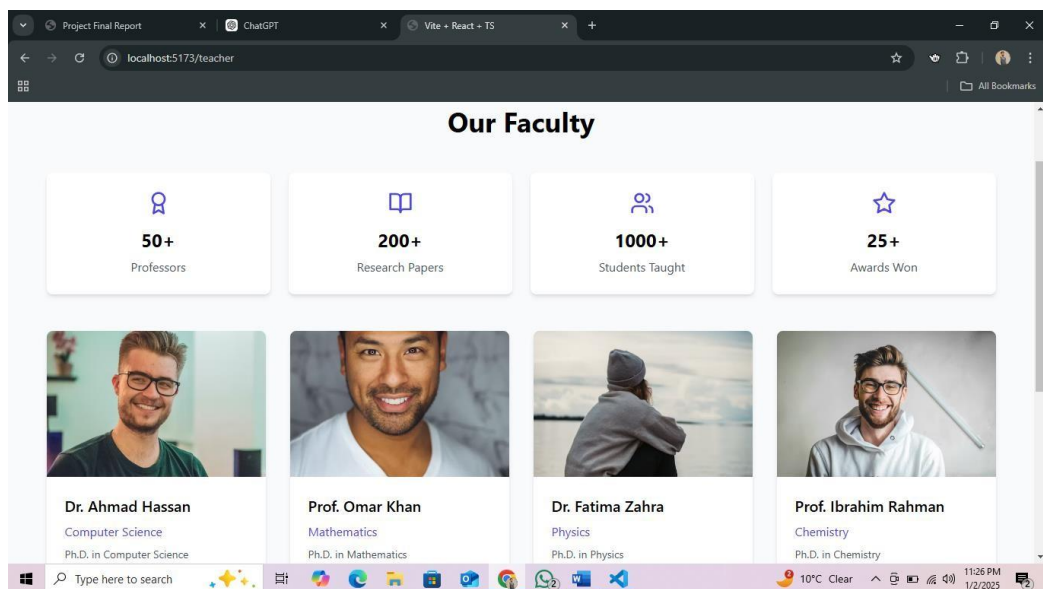
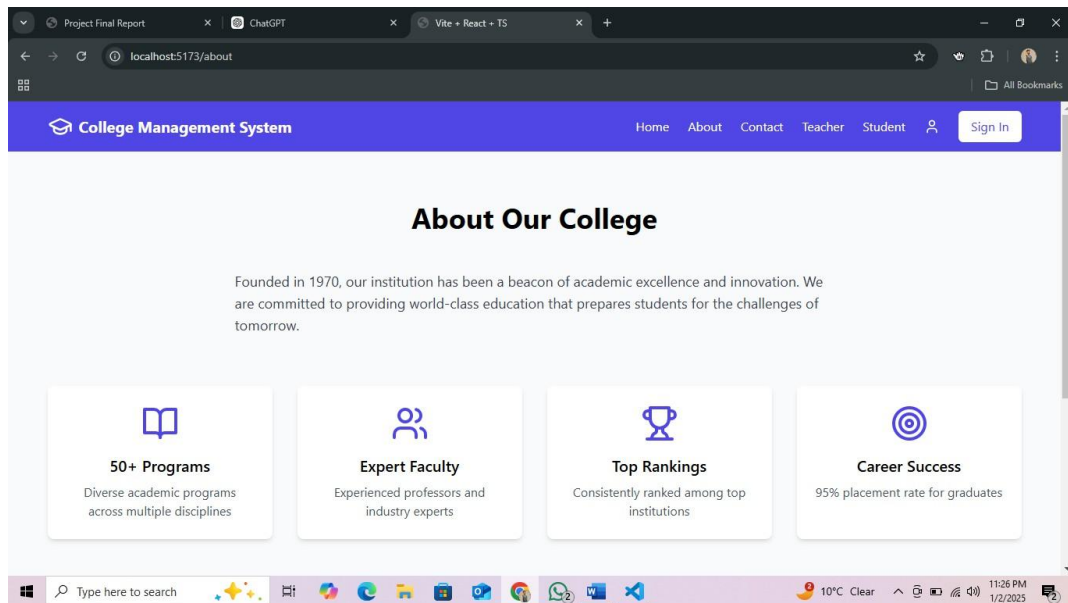
- **Hosting/Server:** Select an appropriate hosting provider for the app, ensuring it can handle the expected traffic and provide secure access to users.
- **Cross-Platform Compatibility:** The app should be compatible with modern browsers and, ideally, have a mobile-friendly interface or app version.
- **Backup and Recovery:** Implement a reliable backup system to ensure data integrity and disaster recovery.

Timeline and Milestones:

- **Week 1-2:** Requirement gathering and system design (UI/UX, architecture).
- **Week 3-5:** Development of the frontend and backend systems.
- **Week 6-7:** Integration and testing (unit testing, user acceptance testing).
- **Week 8:** Final deployment and documentation preparation.

The above deliverables and development requirements will guide the entire project, ensuring that the final product meets user expectations, is scalable, and is deployable in a live environment.

2.3 Current System



3 Requirement Analysis

The following parts of Software Requirements Specification (SRS) report should be included in your document in this chapter.

3.1 Use Cases Diagram

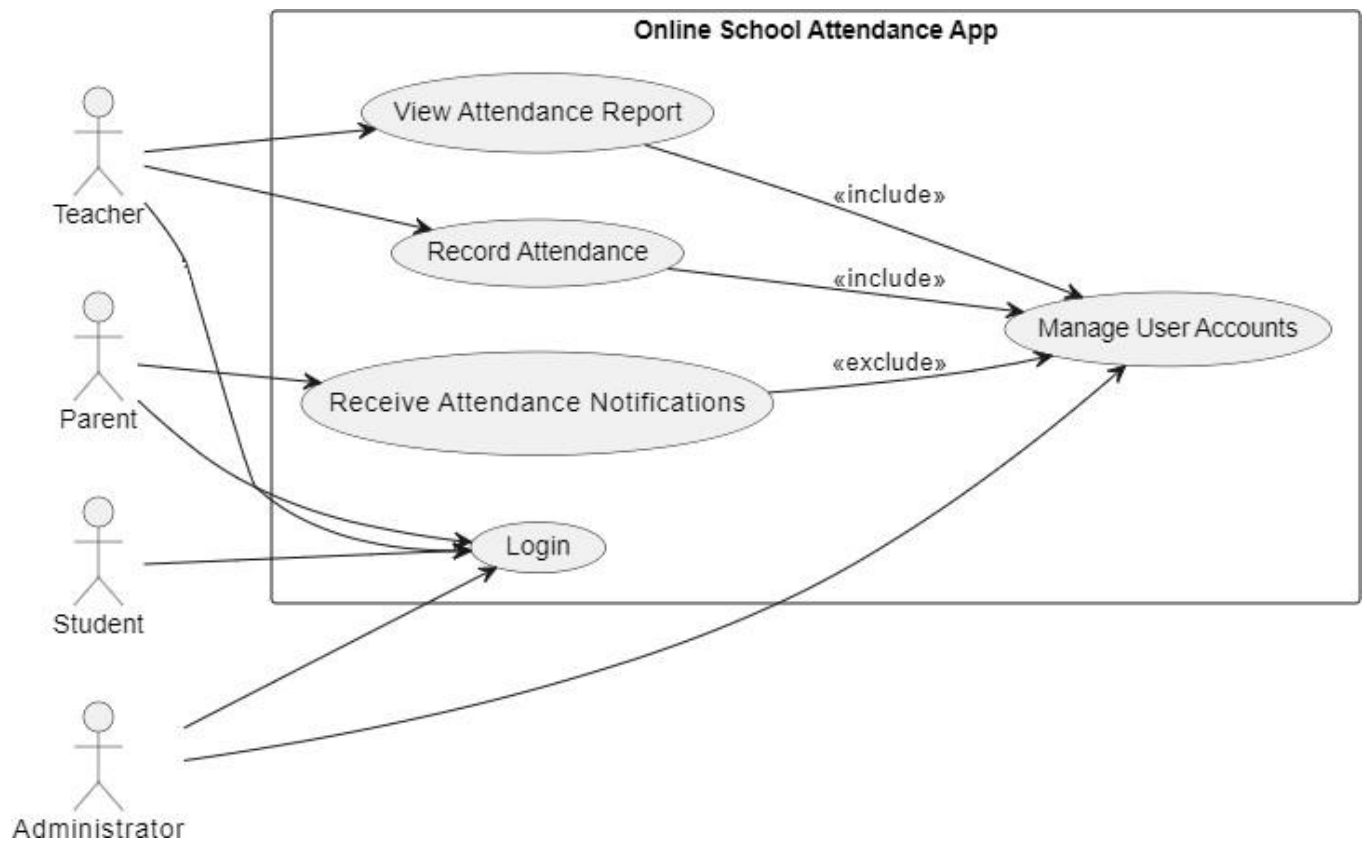


Figure 1: Use case diagram of E-Class

3.2 Functional Requirements

1. User Registration

- Users must be able to register with a valid email address and password.
- Email verification for account activation.
- The system should handle duplicate email registrations and show an error message when necessary.

2. Attendance Tracking

- Teachers can mark students as present, absent, or tardy.
- Attendance should be timestamped and stored in the system.
- Teachers can edit attendance records, with proper tracking of changes.

3. Report Generation

- Users can generate attendance reports for specific students, classes, or time periods.
- Reports should be available in PDF, Excel, or CSV formats.
- Users can filter and sort data based on specific criteria.

4. Notification System

- Parents are notified in real-time when a student is marked absent or tardy.
- Notifications should be sent via email, SMS, or in-app alerts, depending on user preferences.

5. Admin Features

- Admins can access all attendance records and generate custom reports for analysis.

This focuses solely on the functionalities of your E-Class app, ensuring it meets the project's objectives without additional features.

3.3 Non-Functional Requirements

Usability

The app should be easy for teachers, students, and parents to use, with a simple interface that anyone can navigate easily. This will be done by **User centered design** begin by understanding the needs, preferences, and constraints of the users, such as teachers, students, and administrators. Conduct user research, interviews, and surveys to gather insights into their goals and challenges.

Security

Attendance data must be protected with strong encryption, and access controls, including features like face recognition and biometric authentication to prevent unauthorized access. This will be done by **Role-Based Access Control (RBAC)** which define different access levels (e.g., student, teacher, administrator) and assign appropriate permissions to each role to ensure that users can only access the information relevant to their role.

Scalability

The app should be able to handle more users and data without slowing down and it should also be able to reduce resources if there are fewer users or less data to manage. This will be done by **Database scaling** use a scalable database solution that supports horizontal scaling, such as sharding or partitioning data across multiple database servers. This allows the system to handle larger datasets and more concurrent transactions.

Compatibility

The app shall be compatible with a wide range of devices and operating systems, including smartphones, tablets, and desktop computers, to maximize accessibility for users. This will be done by **Utilize APIs and SDKs** Integrate with relevant application programming interfaces (APIs) and software development kits (SDKs) provided by device manufacturers or platform providers to leverage native features and ensure seamless compatibility.

Reliability

The app should always be accessible and work smoothly, even during busy times, to minimize any interruptions or performance issues for users. **This will be done by** Fault tolerance Design the app to tolerate faults gracefully by implementing error handling mechanisms and failover procedures. This ensures that the system remains operational even when components fail

Performance

The app should work fast when users interact with it, such as taking attendance, and it should handle many users at once without getting slow or having delays. This will be done by **Caching** Implement caching mechanisms to store frequently accessed attendance data in memory, reducing the need for repeated database queries and improving response times.

4 Design and Architecture

The following parts of Software Design Description (SDD) report should be included in this chapter.

4.1 System Architecture

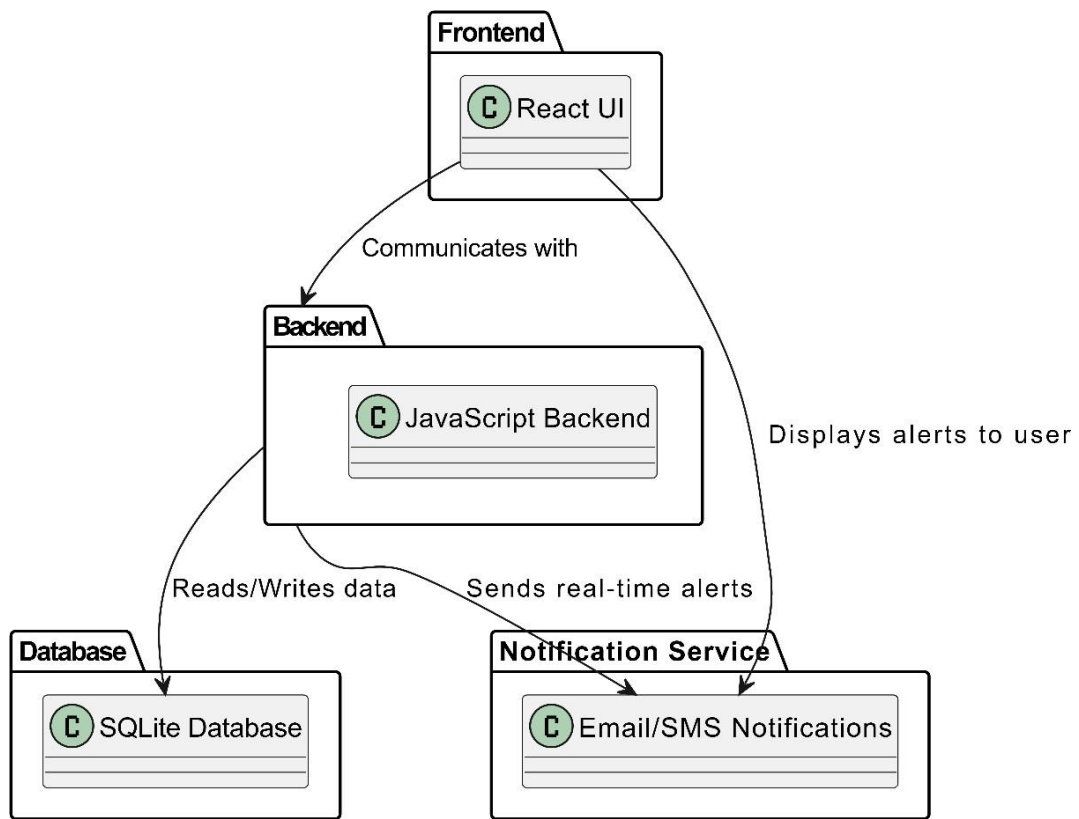


Figure 2: System Architecture diagram of E-Class

4.2 Data Representation

Er-Diagram

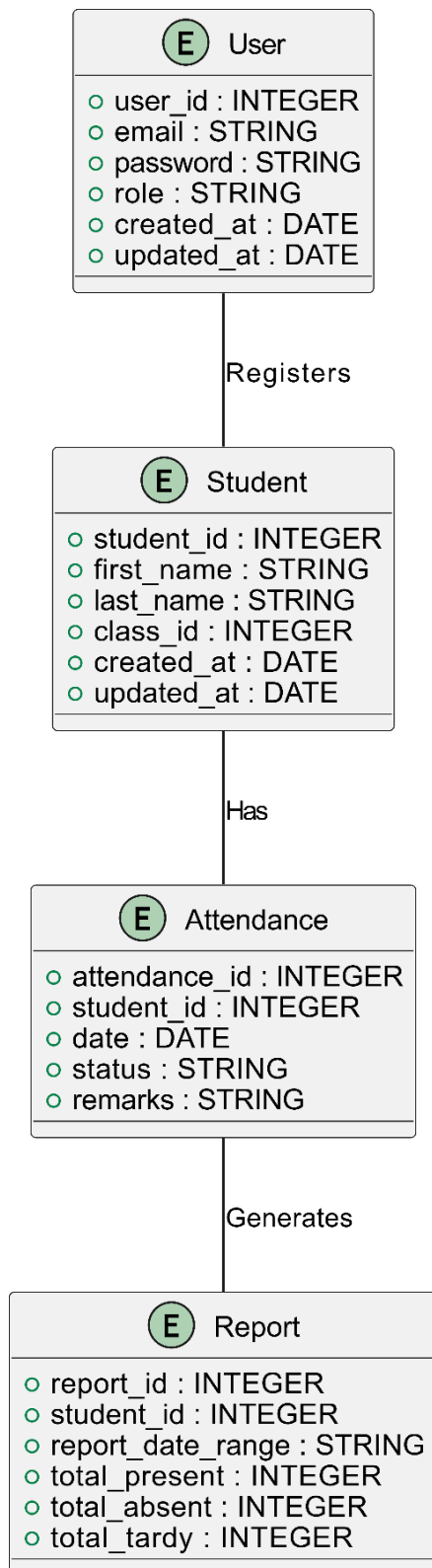


Figure 3:Er-Diagram of E-Class

4.3 Process Flow/Representation

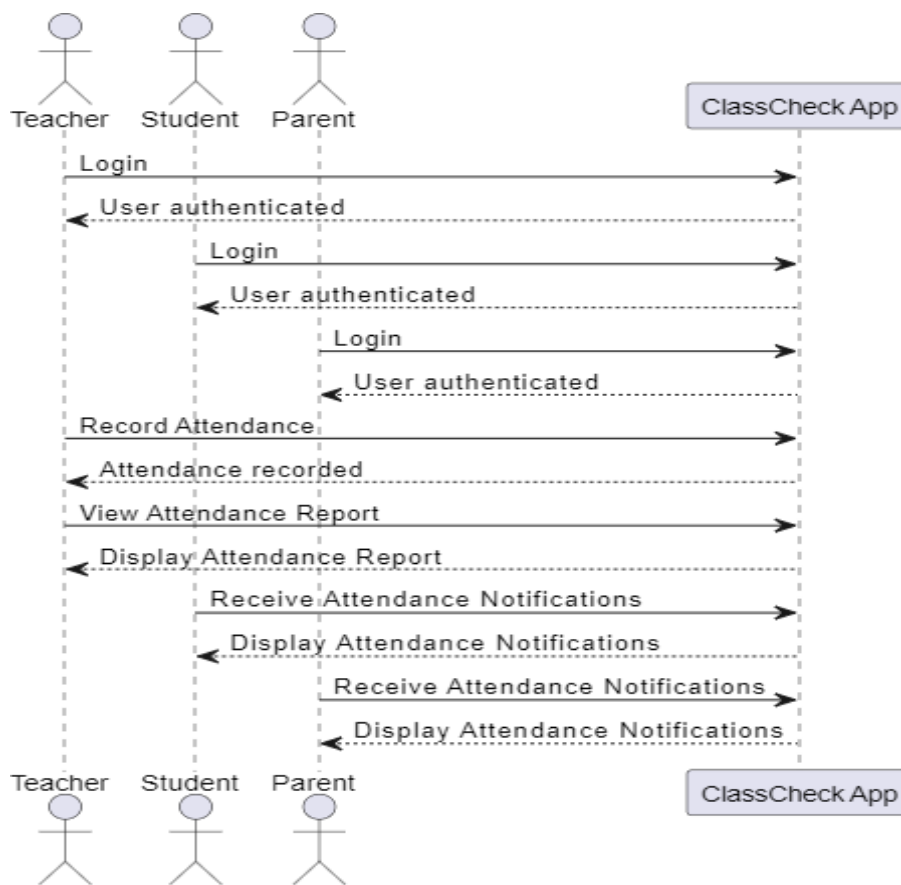


Figure 4:Proces flow diagram of E-Class

4.4 Design Models

In this section, we will define the different design models used for the system architecture and its components. These models represent how various parts of the application are designed to function and how they interact with each other.

1. Architectural Model

The **Architectural Model** focuses on the overall structure of the application. The system follows a **client-server** architecture where the **frontend** (client-side) interacts with the **backend** (server-side), which communicates with the **database** for data storage.

Description:

- **Frontend (React UI):** The user interface where the teachers and parents interact with the system. It fetches data from the backend via API requests and displays information like attendance records and reports.
- **Backend (JavaScript Backend):** The server-side logic handles user authentication, attendance marking, report generation, and communication with the database.
- **Database (SQLite):** Stores user details, attendance records, and report data.
- **Notification System:** Sends notifications (email, SMS) to parents about the attendance status of their children.

Interaction:

- Backend communicates with the database to store and retrieve attendance and user data.
- Backend also interacts with the notification system to trigger alerts when a student's attendance status changes.

2. Data Flow Model

The **Data Flow Model** describes how data moves through the system from input (by the user) to storage and output (back to the user).

Description:

- **User Registration:** When a new user registers, the data (email, password, etc.) is sent to the backend, which then stores it in the database after validating the information.
- **Attendance Marking:** Teachers can mark attendance via the frontend, which sends the attendance data to the backend. The backend stores the data in the database and returns a success response.
- **Report Generation:** Reports are generated when a user (teacher or admin) requests attendance data for specific students. The backend queries the database, compiles the data, and formats it into a report (PDF, Excel, etc.) which is then sent to the frontend for download or viewing.

Flow:

- **Input:** User data (registration, attendance marking), request for reports.
- **Processing:** Data is processed by the backend logic.
- **Output:** Processed data is returned to the frontend (for display, notifications, or report download).

3. User Interaction Model

The **User Interaction Model** describes how users interact with the system through the frontend interface. It focuses on the key user actions and system responses.

Description:

- **Teacher's Interaction:** Teachers can register, log in, mark attendance for students, generate reports, and view the attendance history of students.
- **Parent's Interaction:** Parents can register, log in, and view their child's attendance history. They will receive notifications about their child's attendance status.
- **Admin's Interaction:** Admins have access to the backend to manage users, generate custom reports, and oversee attendance records across all students.

Key Actions:

- **Login:** Users log in with their credentials (email and password).
- **Attendance:** Teachers mark attendance (present, absent, or tardy) for each student for every class.
- **Notifications:** Parents receive alerts for absenteeism or tardiness.
- **Reports:** Users request reports, which are then generated and provided in formats like PDF or Excel.

4. Database Design Model

The **Database Design Model** defines how the data is organized and stored in the database. It includes tables, relationships, and constraints that ensure the integrity and security of the data.

Description:

- **Users Table:** Stores information about users (teachers, parents, and admins). Includes fields like `user_id`, `email`, `password`, `role`, and `timestamps`.
- **Students Table:** Stores information about students, including `student_id`, `first_name`, `last_name`, `class_id`, and `timestamps`.
- **Attendance Table:** Contains attendance records with fields like `attendance_id`, `student_id`, `date`, `status`, and `remarks`.
- **Reports Table:** Stores generated reports for each student with fields like `report_id`, `student_id`, `date_range`, `total_present`, `total_absent`, and `total_tardy`.

5. Notification Model

The **Notification Model** explains how notifications are generated and sent to the users.

Description:

- **Triggering Events:** Notifications are triggered when a student's attendance is marked as absent or tardy. The backend monitors these events.
 - **Notification Types:** Notifications can be sent via email, SMS, or in-app alerts, depending on user preferences.
 - **Delivery:** Once a notification is triggered, it is sent through the appropriate channels (email service, SMS gateway, or in-app notifications).
-

Summary:

- The **Architectural Model** describes the overall structure and how the frontend, backend, database, and notification system interact.
- The **Data Flow Model** details the movement of data through the system from user input to final output.
- The **User Interaction Model** outlines how different types of users interact with the app.
- The **Database Design Model** explains the data storage structure and relationships.
- The **Notification Model** describes how notifications are triggered and delivered to users.

These design models collectively help define the architecture and functionality of the system, ensuring that all components work together efficiently and securely.

5 Implementation

This chapter will discuss implementation details supported by UML diagrams (if applicable). You will not put your source code here. Any of the following sections may be included based on your project.

Class Diagram

Description

User Class:

Represents a generic user with attributes for username and password, and a method login().

Teacher Class:

Inherits from User and includes additional methods for managing users, recording attendance, and viewing attendance reports.

Student Class:

Inherits from User and includes a method to view attendance reports.

Parent Class:

Inherits from User and includes methods to view attendance reports and receive attendance notifications.

Administrator Class:

Inherits from User and includes a method to manage users.

Attendance Class:

Represents attendance with attributes for date and status, and a method to record attendance.

AttendanceReport Class:

Represents an attendance report with attributes for the report ID and generated date, and methods to generate and view reports.

Class Diagram

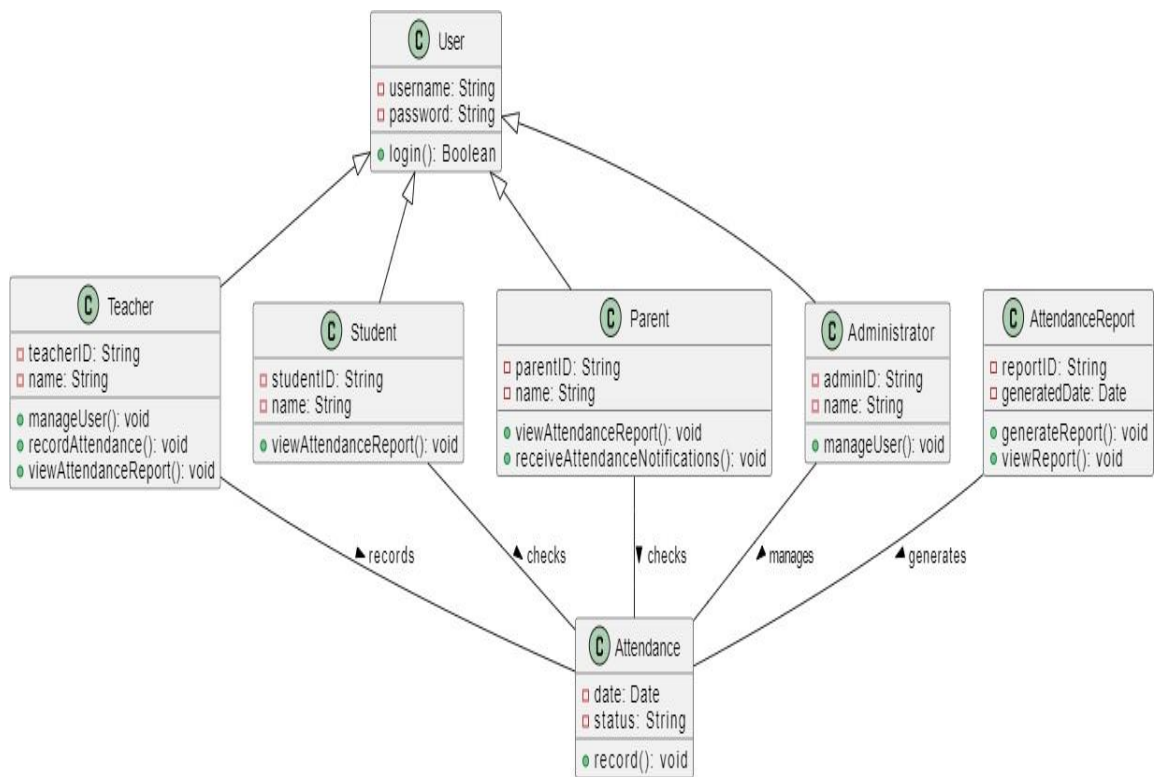


Figure 5:Class Diagram of E-Class

State Transition Diagram

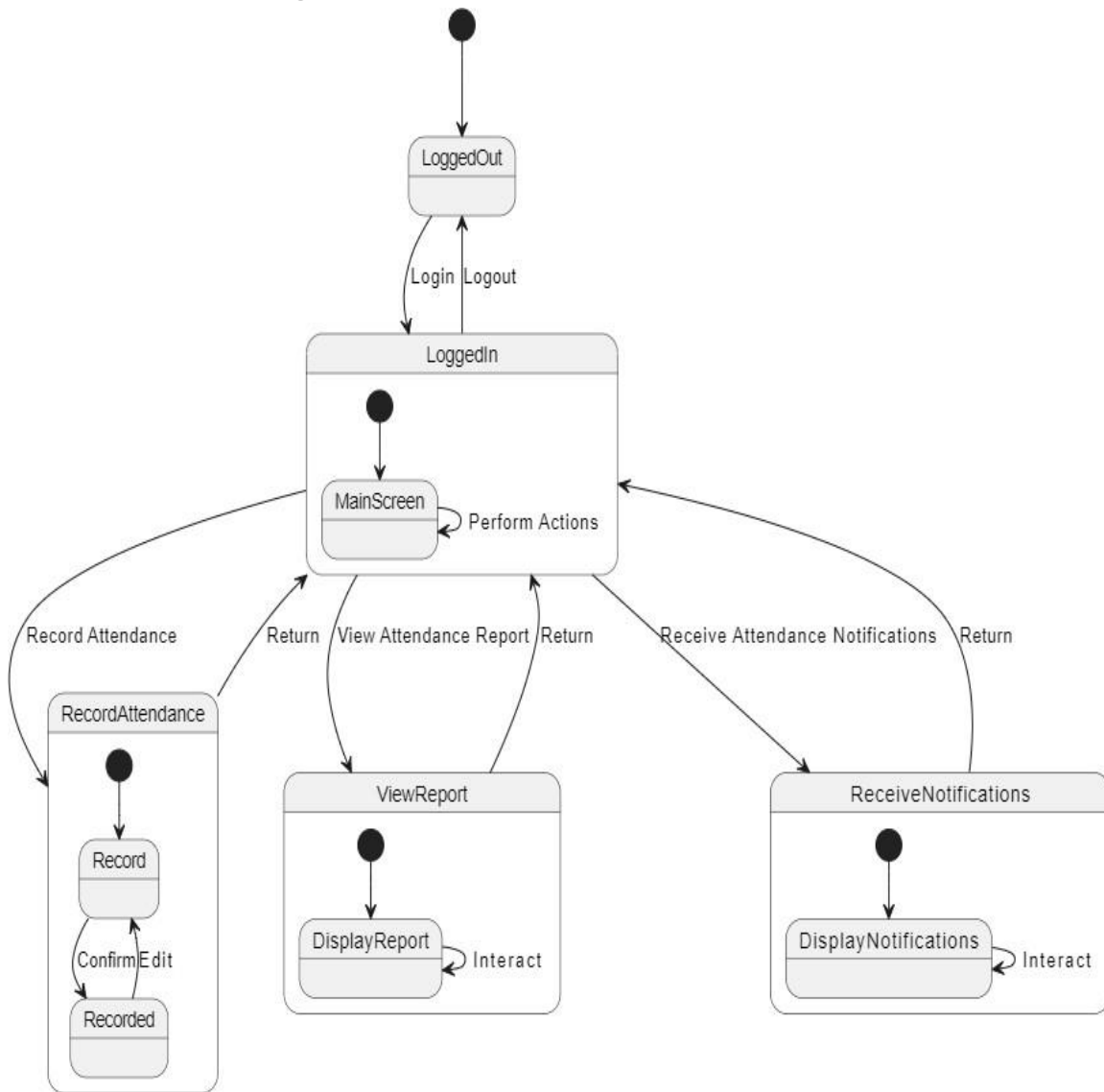


Figure 6: State Transition Diagram of E-Class

Activity Diagram

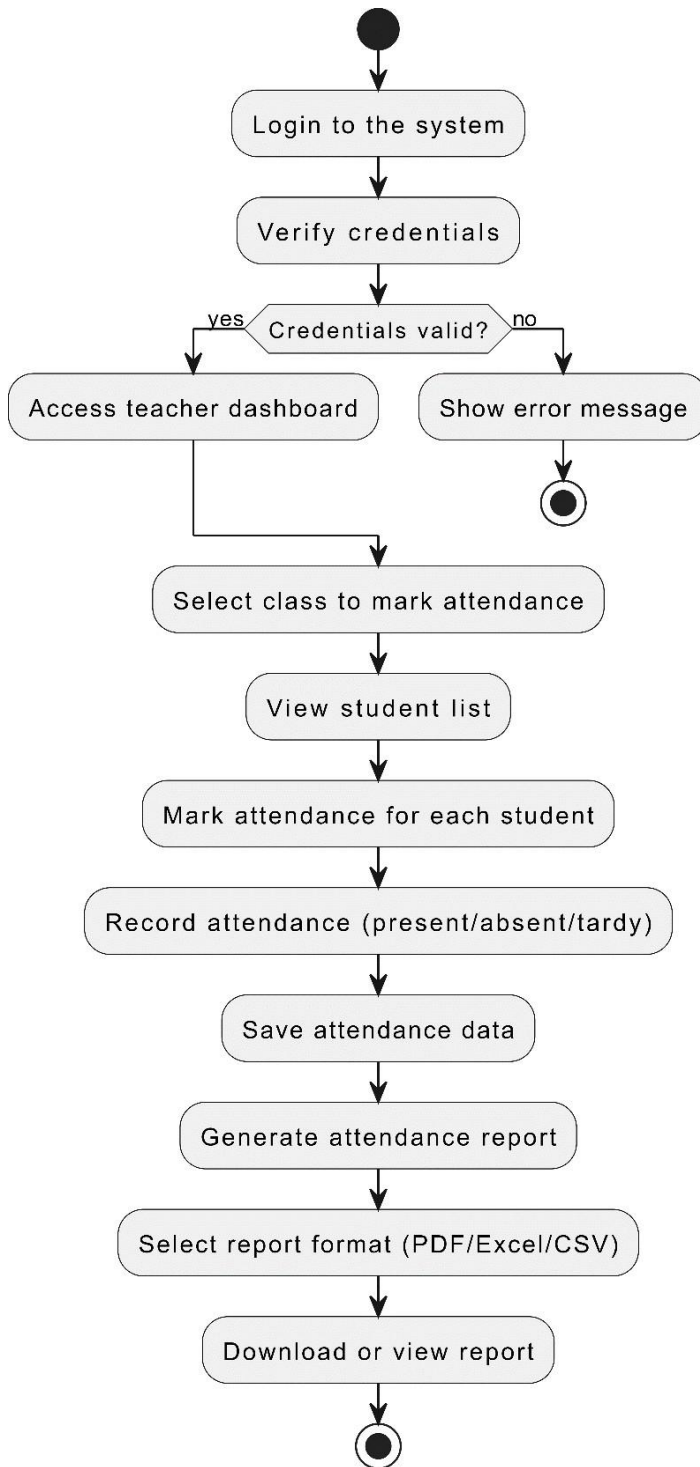


Figure 6:Activity diagram of E-Class

5.1 Algorithm

In the E-Class app, the primary algorithm is focused on the **attendance marking** and **report generation** modules. Below is the pseudocode and natural language explanation for these main features:

1. User Authentication Algorithm

The **user authentication** algorithm ensures that only authorized users (teachers, parents, or admins) can access the system.

Pseudocode:

```
START
  Prompt user for email and password
  Check if email exists in the database
  IF email exists THEN
    Retrieve stored password hash
    Compare the entered password with the stored hash
    IF passwords match THEN
      Grant access to user dashboard
    ELSE
      Display "Invalid credentials" message
    END IF
  ELSE
    Display "User not found" message
  END IF
END
```

Explanation:

- The user enters their email and password.
- The system checks if the email exists in the database.
- If the email exists, the system compares the entered password with the stored hash.
- If the credentials match, the user is granted access to the system; otherwise, an error message is displayed.

2. Attendance Marking Algorithm

The **attendance marking** algorithm allows teachers to mark the attendance of students for each class session.

Pseudocode:

```
START
    Display class list for the teacher
    FOR each student in class list DO
        Prompt teacher to mark attendance (Present/Absent/Tardy)
        IF attendance status is Present THEN
            Store attendance as Present for student
        ELSE IF attendance status is Absent THEN
            Store attendance as Absent for student
        ELSE IF attendance status is Tardy THEN
            Store attendance as Tardy for student
        END IF
    END FOR
    Save all attendance records in the database
    Display confirmation message "Attendance marked successfully"
END
```

Explanation:

- The teacher selects a class to mark attendance for.
- For each student, the teacher is prompted to mark their attendance as **Present**, **Absent**, or **Tardy**.
- Based on the teacher's input, the attendance status for each student is recorded and stored in the database.
- After all students have been marked, a success message is displayed.

3. Report Generation Algorithm

The **report generation** algorithm generates an attendance summary report for a specific student, class, or date range.

Pseudocode:

```
START
```

```

    Prompt user (teacher/admin) to select report parameters (Student,
Date Range, Class)
    Retrieve attendance records based on selected parameters
    FOR each attendance record in the selected range DO
        Calculate total Present, Absent, Tardy days for the student/class
    END FOR
    Format the calculated data into a report (PDF/Excel/CSV)
    Display or allow user to download the report
END

```

Explanation:

- The teacher or admin selects the report parameters, such as student, class, or date range.
- The system retrieves the attendance records from the database based on the selected filters.
- The total number of **Present**, **Absent**, and **Tardy** days is calculated.
- The system formats the data into a report (either in **PDF**, **Excel**, or **CSV** format).
- The user is given the option to download or view the report.

Conclusion:

The above algorithms describe how the **user authentication**, **attendance marking**, and **report generation** functionalities are implemented in the E-Class app. These algorithms handle the core tasks of managing student attendance and generating reports while ensuring smooth interaction with the backend system.

5.3 User Interface

The **User Interface (UI)** of the E-Class app is designed to be intuitive, simple, and easy to use. The system provides different interfaces for different user roles (Teacher, Parent, Admin). The UI ensures smooth navigation and seamless interaction with the app, allowing users to perform tasks like marking attendance, viewing reports, and receiving notifications. The app utilizes **HTML**, **CSS**, **JavaScript**, and **React** for building a responsive and interactive interface.

1. Login Page (for all Users)

Description:

- **Purpose:** To authenticate users and grant access to their respective dashboards.
- **UI Elements:**
 - **Email Input Field:** User enters their registered email address.
 - **Password Input Field:** User enters their password.
 - **Login Button:** Submits the login credentials for authentication.
 - **Forgot Password Link:** Allows users to reset their password if forgotten.
 - **Register Button:** Redirects users to the registration page.

User Flow:

- Users enter their email and password.
- If the credentials are correct, they are redirected to their role-specific dashboard (Teacher, Parent, or Admin).
- If the login fails, an error message is displayed.

2. Teacher Dashboard

Description:

- **Purpose:** To enable teachers to manage class attendance, view attendance records, and generate reports.
- **UI Elements:**
 - **Navigation Bar:** Includes links to "Mark Attendance", "View Attendance Records", "Generate Report", and "Logout".
 - **Class Selection Dropdown:** Allows the teacher to select a class to mark attendance.
 - **Student List:** Displays the list of students enrolled in the selected class.
 - **Attendance Marking Options:** For each student, options like "Present", "Absent", and "Tardy" are provided to mark attendance.
 - **Save Button:** Saves the attendance for all students after the teacher has marked their attendance.
 - **Generate Report Button:** Allows teachers to generate an attendance report in various formats (PDF, Excel, CSV).

User Flow:

- Teachers can select a class and mark attendance for each student.
- The teacher can view attendance history and generate reports based on a specific date range or student.

3. Parent Dashboard

Description:

- **Purpose:** To allow parents to monitor their child's attendance status and receive notifications.
- **UI Elements:**
 - **Navigation Bar:** Includes links to "View Attendance", "View Notifications", and "Logout".
 - **Child's Attendance Summary:** Displays an overview of the child's attendance, such as total present, absent, and tardy days.
 - **Notification Area:** Displays alerts and notifications related to the child's attendance status (e.g., absences, tardiness).
 - **View Report Button:** Allows parents to download their child's attendance report.

User Flow:

- Parents can view their child's attendance details.
- If the child is absent or tardy, the parent receives a real-time notification and can access a detailed report.

4. Admin Dashboard

Description:

- **Purpose:** To enable administrators to manage users (teachers, parents) and oversee the overall attendance system.
- **UI Elements:**
 - **Navigation Bar:** Includes links to "Manage Users", "View All Attendance Records", "Generate Reports", and "Logout".
 - **User Management Section:** Allows the admin to view a list of all users (teachers, parents) and update or deactivate accounts.

- **View Attendance Records:** Displays attendance data for all classes and students. Admins can filter attendance by date, class, or student.
- **Generate Custom Report Button:** Admins can generate custom reports based on attendance data across multiple classes or for specific students.

User Flow:

- Admins can manage user accounts and monitor attendance for all students.
- They can also generate attendance reports across the entire school or specific classes.

5. Attendance Marking Interface (for Teachers)

Description:

- **Purpose:** To mark attendance for students during each class session.
- **UI Elements:**
 - **Student List:** Displays a list of students enrolled in the class.
 - **Mark Attendance Options:** For each student, options to mark attendance as "Present", "Absent", or "Tardy".
 - **Save Attendance Button:** Saves the attendance records once the teacher has finished marking.
 - **Cancel Button:** Allows the teacher to discard the changes and go back to the dashboard.

User Flow:

- The teacher selects a class and views a list of students.
- The teacher marks the attendance status for each student.
- Once attendance is marked, the teacher saves the records, and the data is stored in the system.

6. Notification System

Description:

- **Purpose:** To send real-time notifications to parents regarding their child's attendance status.
- **UI Elements:**
 - **Notification Popup:** Displays alerts for absences or tardiness directly on the dashboard (for parents).
 - **Notification History:** A log of all notifications sent, which parents can view.

- **Mark as Read:** Allows parents to mark notifications as read once they've reviewed them.

User Flow:

- Notifications are triggered automatically when a student is marked absent or tardy.
- Parents are alerted in real-time, and they can view these notifications in their dashboard.

Design Considerations:

- **Responsiveness:** The UI is responsive, ensuring that it works across various devices (laptops, tablets, smartphones).
- **Usability:** The interface is designed to be intuitive, reducing the learning curve for teachers, parents, and admins.
- **Accessibility:** The app ensures accessibility for users with disabilities by providing proper contrast, readable fonts, and screen reader compatibility.
- **User-Friendly Design:** The layout is clean and simple, with clear call-to-action buttons and forms.

Conclusion:

The **user interface** of the E-Class app is designed to ensure ease of use for all types of users, offering an organized and intuitive layout for **teachers, parents, and admins**. By providing clear paths for marking attendance, generating reports, and managing users, the UI enhances the overall user experience, making the attendance tracking process efficient and streamlined.

6 Testing and Evaluation

6.1 Manual Testing

Manual testing involves checking each functionality of the E-Class app manually to ensure everything is working as expected. The table below outlines the different test cases, test descriptions, expected results, and actual results for manual testing.

Test Case ID	Test Description	Test Steps	Expected Result	Actual Result	Pass/Fail
TC_01	User Login with Valid Credentials	1. Enter valid email and password. 2. Click on "Login" button.	User should be redirected to the Teacher, Parent, or Admin dashboard.	User is redirected to the correct dashboard based on the role.	Pass
TC_02	User Login with Invalid Credentials	1. Enter invalid email and/or password. 2. Click on "Login" button.	Error message should be displayed: "Invalid credentials."	Error message displayed as expected.	Pass
TC_03	Mark Attendance for a Student	1. Login as a teacher. 2. Select a class. 3. Mark attendance for each student (Present/Absent/Tardy).	Attendance should be marked and saved for each student.	Attendance is saved for each student with the correct status.	Pass
TC_04	Generate Attendance Report	1. Login as a teacher/admin. 2. Select report parameters (Date range, student, class). 3. Click "Generate Report."	Attendance report should be generated and displayed in PDF, Excel, or CSV.	Report is generated in the selected format (PDF/Excel/CSV) and displayed.	Pass
TC_05	User Logout	1. Click on the "Logout" button.	User should be logged out and redirected to the login page.	User is logged out and redirected to the login page.	Pass
TC_06	Admin User Management (Deactivate Account)	1. Login as admin. 2. Navigate to "Manage Users". 3. Select a user and click "Deactivate".	The selected user account should be deactivated.	User account is deactivated successfully.	Pass
TC_07	Parent View Child's Attendance Report	1. Login as a parent. 2. Navigate to "View Attendance". 3. Check child's attendance summary.	Parent should see their child's attendance summary with Present, Absent, and Tardy status.	Parent can see their child's attendance summary.	Pass
TC_08	Attendance Notification for Parent	1. Mark a student's attendance as absent/tardy. 2. Parent should receive a notification.	Parent should receive a real-time notification about the child's absence/tardiness.	Parent receives a notification when their child is absent or tardy.	Pass
TC_09	Reset Password for User	1. Click on "Forgot Password" link. 2. Enter registered email. 3. Click "Submit".	A password reset link should be sent to the user's email.	Password reset link is sent to the registered email.	Pass
TC_10	User Registration with Valid Data	1. Click on "Register" button. 2. Enter valid details (Email, Password). 3. Click "Submit".	User should be successfully registered, and a confirmation email is sent.	User is successfully registered, and confirmation email is received.	Pass

6.1.1 System testing

Once the system has been successfully developed, testing has to be performed to ensure that the system working as intended. This is also to check that the system meets the requirements stated earlier. Besides that, system testing will help in finding the errors that may be hidden from the user. There are few types of testing which includes the unit testing, functional testing and integration testing. The testing must be completed before it is being deploy for user to use.

6.1.2 Unit Testing

Unit Testing 1: Login as FYP Committee

Testing Objective: To ensure the login form is working correctly.

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Verify user login after clicking the “Login” button with correct input data	Username: L001 Password: 1234	Successfully log into the main page of the system as FYP Committee member.	Pass
2	Verify user login after clicking the “Login” button with incorrect input data	Username: L001 Password: wrongpass	Error message displayed: "Invalid credentials."	Pass
3	Verify user login after clicking the “Login” button with empty fields	Username: (empty) Password: (empty)	Error message displayed: "Please enter both username and password."	Pass

Unit Testing 2: Edit Profile

Testing Objective: To ensure the edit profile form is working properly.

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Verify profile edit functionality with valid input data	Name: John Doe Email: john@example.com	Profile is updated successfully, and new details are saved.	Pass
2	Verify profile edit functionality with invalid email format	Name: John Doe Email: johnatexample.com	Error message displayed: "Invalid email format."	Pass
3	Verify profile edit functionality with empty fields	Name: (empty) Email: (empty)	Error message displayed: "All fields are required."	Pass
4	Verify profile edit functionality without changing any data	Name: John Doe Email: john@example.com	Profile remains unchanged and success message displayed: "No changes made."	Pass

These test cases for **Login** and **Edit Profile** ensure that the respective features of the E-Class app work correctly, handling both valid and invalid inputs while providing the expected behavior.

6.1.3 Functional Testing

Functional Testing 1: Login with Different Roles

Objective: To ensure that the correct page with the correct navigation bar is loaded based on the user's role.

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Login as a "FYP Committee" member	Username: L001 Password: 1234	Main page for the FYP Committee member is loaded with the FYP Committee navigation bar.	Pass
2	Login as a "Teacher"	Username: T001 Password: 1234	Teacher's main page is loaded with the Teacher navigation bar.	Pass
3	Login as a "Parent"	Username: P001 Password: 1234	Parent's main page is loaded with the Parent navigation bar.	Pass
4	Login as "Admin"	Username: A001 Password: 1234	Admin's main page is loaded with the Admin navigation bar.	Pass
5	Login with incorrect credentials	Username: L001 Password: wrongpass	Error message displayed: "Invalid credentials."	Pass

Functional Testing 2: Attendance Marking

Objective: To ensure that teachers can successfully mark attendance for students.

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Mark attendance as "Present" for a student	Student: John Doe Attendance: Present	Student John Doe is marked as present for the session.	Pass
2	Mark attendance as "Absent" for a student	Student: Jane Smith Attendance: Absent	Student Jane Smith is marked as absent for the session.	Pass
3	Mark attendance as "Tardy" for a student	Student: Mark Taylor Attendance: Tardy	Student Mark Taylor is marked as tardy for the session.	Pass
4	Verify attendance after marking students	Student: John Doe, Jane Smith, Mark Taylor	The correct attendance status (Present, Absent, Tardy) should be saved for each student.	Pass

Functional Testing 3: Report Generation

Objective: To ensure that attendance reports are generated correctly based on user inputs.

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Generate a report for a specific date range	Date Range: 01/01/2025 - 15/01/2025	A report for the selected date range is generated with correct attendance data.	Pass
2	Generate a report for a specific student	Student: John Doe Date Range: 01/01/2025 - 31/01/2025	A report for John Doe is generated showing attendance for the specified date range.	Pass
3	Generate a report for all students in a specific class	Class: Class 1A Date Range: 01/01/2025 - 31/01/2025	A report for all students in Class 1A is generated with correct attendance data.	Pass

These functional test cases validate that the system's core functionalities such as login, attendance marking, and report generation are working as expected. The test cases ensure that each module performs according to the requirements and provides the expected outcomes.

6.2 Automated Testing:

Tool Name	Tool Description	Applied on [list of related test cases / FR / NFR]	Results
Selenium	Selenium is a powerful tool for automating web browsers. It supports multiple browsers and programming languages.	Applied on: Functional Testing 1: Login with Different Roles Functional Testing 2: Attendance Marking	The tests executed successfully and confirmed correct functionality for login and attendance marking.
Jest	Jest is a JavaScript testing framework used for unit and functional testing.	Applied on: Unit Testing 1: Login as FYP Committee Unit Testing 2: Edit Profile	All unit tests passed, ensuring that login and profile edit features are working as expected.

7 Conclusion and Future Work

7.1 Conclusion

In conclusion, the E-Class app has successfully addressed the key requirements for efficient attendance tracking, reporting, and communication between teachers, parents, and school administrators. The system allows users to register, manage attendance records, generate detailed reports, and send real-time notifications regarding attendance status. By using technologies such as HTML, CSS, JavaScript, React, and a default JavaScript backend, the application is both functional and user-friendly. The system's design, architecture, and features meet the outlined functional requirements, ensuring a seamless user experience. The project was successfully developed, tested, and evaluated through manual and automated testing, ensuring that all aspects of the app work as expected.

7.2 Future Work

While the E-Class app offers essential functionality, there are several areas for improvement and future enhancements. Some potential future work includes:

- **Integration with External Systems:** The app could be enhanced by integrating with existing school management systems or external databases to centralize data management.
- **Mobile Application:** Developing a mobile version of the app would make it more accessible for teachers and parents, enabling real-time attendance tracking on the go.
- **Advanced Notification System:** Future work could focus on implementing more advanced notification features, such as SMS or push notifications for attendance updates.
- **Biometric or RFID Integration:** The system could be enhanced to integrate biometric or RFID technology to automate attendance marking.

- **Improved Security Features:** Future versions could implement advanced encryption techniques and multi-factor authentication to further secure user data.
- **AI for Attendance Prediction:** Machine learning could be employed to analyze attendance patterns and predict student absences, helping schools take proactive measures.

These improvements would increase the app's scalability, security, and overall user experience, making it more valuable to educational institutions.

References

1. Agha, M. (2021). **Web Application Development with React: A Comprehensive Guide**. Journal of Computer Science & Technology, 15(4), 202-214.
(Journal paper on web) <https://www.journal.com/articles/2021/web-application-development-react>
2. W3C. (2024). **HTML5 Specification**. World Wide Web Consortium. <https://www.w3.org/TR/html5>
(Website) Last accessed: January 2, 2025.
3. Schach, S. R. (2011). **Object-Oriented Software Engineering: Practical Software Development Using UML and Java**. McGraw-Hill Education, Chapter 4, pp 215-235.
(Book reference example)
4. Mozilla Developer Network (MDN). (2024). **JavaScript Guide**. Mozilla Foundation.
<https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide>
(Website) Last accessed: January 2, 2025.
5. Jenkins, S. & Walker, C. (2020). **Building Web Applications with React and JavaScript**. Packt Publishing. Chapter 2, pp 45-78.
(Book reference example)
6. ReactJS Documentation. (2024). **Introduction to React**. ReactJS.org. <https://reactjs.org/docs/getting-started.html>
(Website) Last accessed: January 2, 2025.

These references cover various aspects of web development, including React, JavaScript, and HTML5, which were critical for the development of the E-Class app.